

Enterprise Cybersecurity Architecture Specification

Strategic Governance and Security Controls for Research (PROD-DSDL) Isolation

1. Executive Summary

This white paper defines the mandatory security control framework for the PROD-DSDL (Data Science and Research) environment within the shared Enterprise Tech Shared Services (ETSS) APIGEE Production instance. The primary objective of this specification is to prevent Research workloads from impacting the stability and security of core business applications (PROD-SFDL, PROD-SFBU).

2. Design Philosophy: The Walled Garden

The core requirement of this architecture is the 'Walled Garden' principle. Any API created within the PROD-DSDL account must be consumed exclusively by entities within the same PROD-DSDL account via the APIGEE Gateway. This creates a logical loopback where the Gateway acts as a non-bypassable security checkpoint.

3. Method-Level Policing and Data Flow

To maintain production data integrity, cross-account communication is restricted to 'Read-Only' patterns.

3.1 Cross-Account GET Restriction

Traffic originating from PROD-DSDL destined for PROD-SFBU or PROD-SFDL is strictly limited to HTTP GET requests. APIGEE must block all POST, PUT, DELETE, and PATCH verbs for these routes to prevent unauthorized data modification.

3.2 Pull-Only Model

Research environments are permitted to pull data from other PROD lifecycle accounts but are prohibited from pushing data or triggering state changes in business applications.

4. Control ID: AC-01 | Network-Layer Security

Northbound ingress is restricted to the authorized CIDR ranges of the PROD-DSDL AWS VPC. Southbound egress is restricted to ETSS APIGEE CIDRs via Producer Security Groups.

5. Control ID: AC-02 | Identity and Entitlement Management

OAuth2 tokens for DSDL must use unique 'research:*' namespaces. Developer Apps must be tagged with a 'tenant_id' to prevent cross-account pivoting.

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6. Control ID: AC-03 | Resource Guardrails

Mandatory Spike Arrest and 10MB payload size limits are enforced for all Research traffic to protect shared resources.

7. Escalation Trigger

If PROD-DSDL traffic exceeds 20 percent of aggregate Gateway utilization, a mandatory migration to a dedicated environment or standalone instance is triggered.